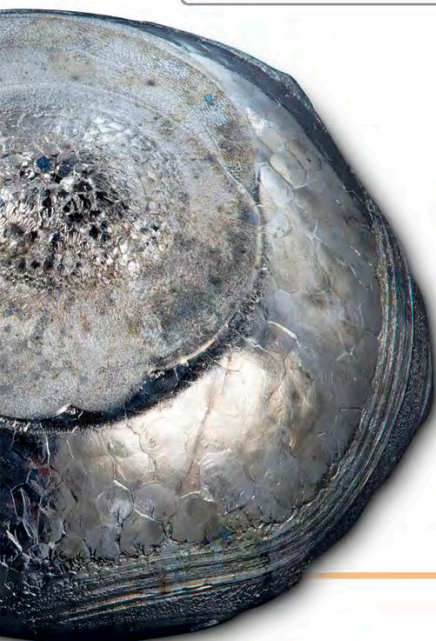
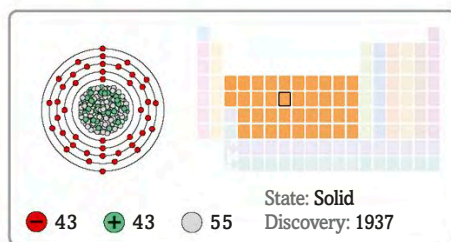


43 Tc Technetium



This experimental sports car is built with a rust-resistant alloy that contains molybdenum.



are hard enough for making sturdy tools, too. Molybdenum alloys are used in the latest designs of supercars, such as the **Vencer Sarthe**.

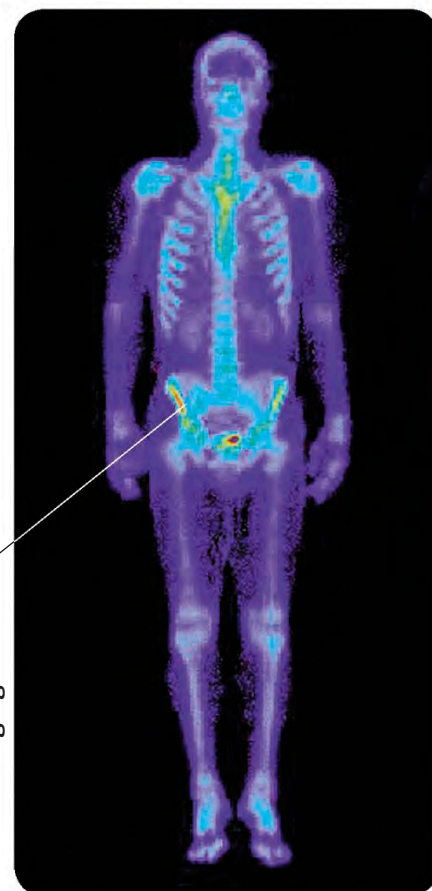


This pure form of the metal is produced inside nuclear reactors.

This body scan was created using the radioactive effects of technetium.

Foil of pure technetium produced in a reactor

Technetium-based imaging



This box contains radioactive molybdenum, which breaks down into technetium.

Generating technetium



Technetium was the first element to be produced artificially by researchers. It is named after the Greek word for artificial, *tekhnētos*. Technetium does not exist in nature: any of its atoms that once existed on Earth broke down millions of years ago. Tiny amounts of this element were discovered in the waste produced by early nuclear reactors. Technetium is the lightest radioactive element. It is used extensively in **medical imaging**. It is injected into a patient's body, where it emits radiation for a short while. Some machines use this radiation to show bones clearly.